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DEEMED TO BE UNIVERSITY

ZELIA

ZERO-LITERACY ELIGIBILITY & LIVELIHOOD INTELLIGENCE ASSISTANT

Presented By:

Group C15

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LITERATURE REVIEW

02

TITLE	CORE CONTRIBUTION	RELEVANCE TO ZELIA	DRAWBACKS
LayoutLM (Xu et al., 2020)	Joint learning of text + layout + visual features for document understanding	Enables accurate form-field extraction from government forms	Limited support for multilingual / handwritten / noisy scans
LayoutXLM (Xu et al.),2020	Extends LayoutLM to multilingual visually-rich documents	Works across Indian regional-language forms	Needs large multilingual data; weaker in low-resource scripts
VisualMRC (Tanaka et al.),2019	Models document understanding as QA / reading comprehension	Helps with explanatory queries (e.g., rejection reasons)	Not suitable for structured form extraction automation
PubLayNet (Zhong et al.),2018	Large-scale dataset for document layout detection	Improves layout segmentation before OCR + field mapping	Dataset biased to scientific papers, not government forms

LITERATURE REVIEW

03

TITLE	CORE CONTRIBUTION	RELEVANCE TO ZELIA	DRAWBACKS
DocVQA (Mathew et al.),2022	Dataset for Visual Question Answering on documents	Useful for future conversational document queries	Less relevant for end-to-end eligibility / form workflows
LayoutLLM (Fujitake),2020	Integrates LLMs with layout-aware reasoning	Potential next-gen adaptive document pipeline	Experimental, high compute, limited real-world validation
Visual Conversational Interfaces (Agarwal et al.),2021	Designs voice + visual workflows for low-literacy users	Guides voice-first UX for CHWs & beneficiaries	Does not address document automation / backend logic
Speech-Based Mobile Interface (Yammiyavar & Gangopadhyay),2023	Usability study of speech-driven mobile interaction	Supports error-tolerant speech UX design	No integration with document or AI-based decision systems

DATASETS USED

Open Government

Public welfare & scheme datasets available

Social Welfare Scheme Resources

Inputs to eligibility rule databases.

NFHS-5

Household literacy, health, and assets data

Master Data: Scheme Funds Released

Prioritize high-impact schemes for recommendation layers.

SECC

Socio-economic eligibility indicators for analysis

MoSPI Socio-Economic Statistics/Surveys

Enrich socio-economic indicators for model features.

Census Tables

Baseline population features and stratification for models.

NDAP

Multi-theme cross-linkable datasets for analytics and machine learning.

State-wise Multidimensional Poverty

Validate eligibility thresholds; calibrate poverty scoring.

04

PROBLEM STATEMENT

60% of eligible rural households never claim their entitled government benefits

- Low Literacy — multi-field paper applications require reading/writing skills most beneficiaries lack
- Complex Procedures — 300+ schemes with varying eligibility criteria, documents, and office routing
- Remote Location — administrative offices far from villages; repeated trips needed per application

05

SOLUTION OVERVIEW

A Voice-First Interface

zero-literacy binary prompts via TFLite ASR, no reading required

B Hybrid Eligibility Engine

deterministic SQL rules + LightGBM ML scorer across 536+ schemes

C PDF Auto-Fill

coordinate-mapped overlay on official government templates, no synthetic pages

D Pictogram Guidance

icon + audio step-by-step task guides for zero-literacy users

E CHW Verification Loop

document-grounded OCR cross-check; voice data tentative, document data authoritative

F Zelia Copilot RAG

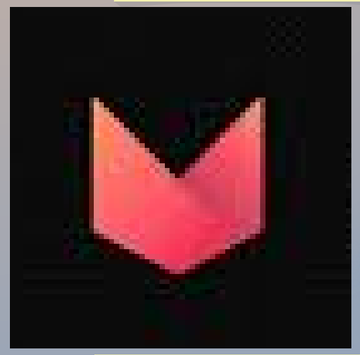
pincecone backend chatbot giving instant scheme answers in natural language

SYSTEM ARCHITECTURE (END TO END PIPELINE)

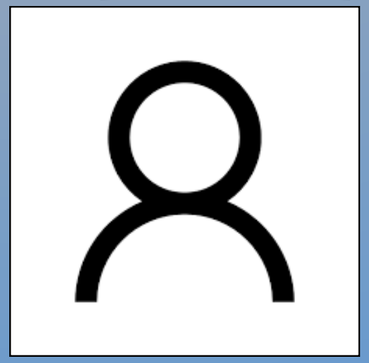
07

metadata stored into database as key value pair

metadata extracted from documents stored as key value pairs in the database



User voice inputs the required details and requests and hash the user id (SHA26)



pdfrw & ReportLab

Precision vector overlay for official govt templates.

pdf refilling service engine

ocr service engine to extract uploaded document contents

n8n workflow orchestration

Webhook

FAST API translates via google/deepl AI

Language processing

DL model to analyze structure of govt templates

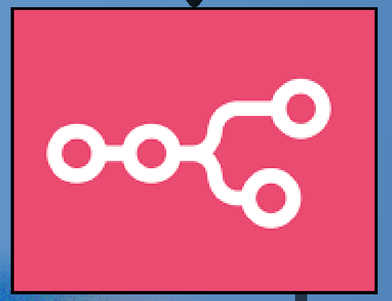
pinecone /rag

LayoutLM

Annoy index

groq to generate response

CHW Resonse sent to frontend



Node orchestration with detailed configs

2 way connection for chw verification to ensure the pdf is filled with required information if not the document is sent to pdf reffiling service which is either accepted or rejected for further submission

filled pdf saved in folder alogn with uploaded documents in uniquely hashed user folder

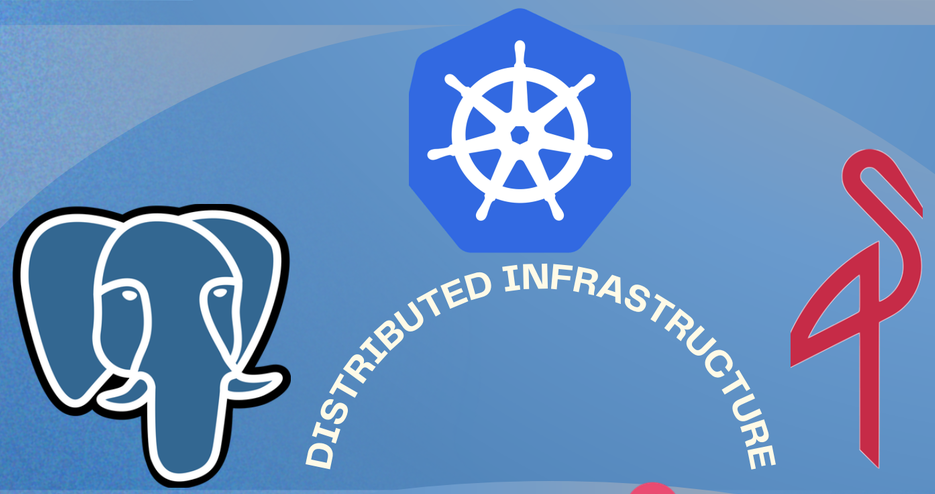
Store

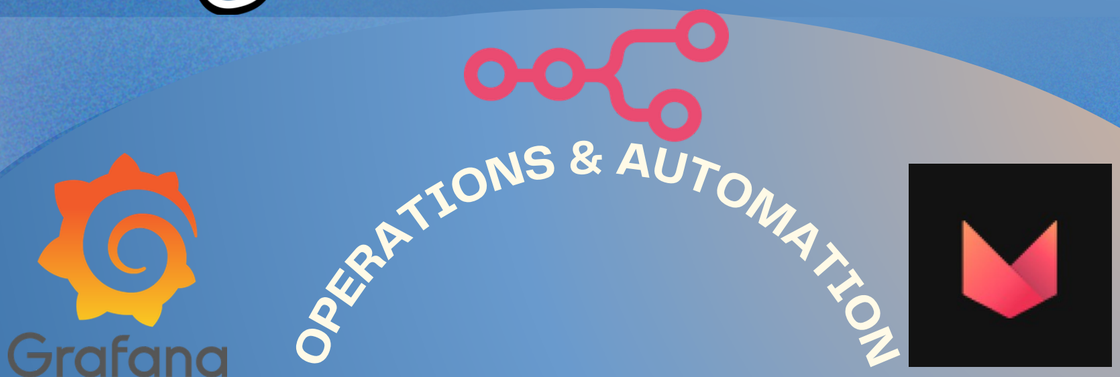
Data

nodes send the user metadata, submissions, logs, schemes

TECHNOLOGY STACK

09





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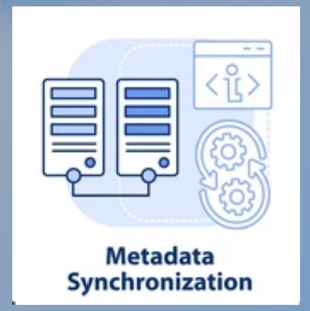
ELIGIBILITY ENGINE

[[0, 1, 0, 0, 0]
[0, 0, 0, 0, 1]
[0, 0, 1, 0, 0]
[1, 0, 0, 0, 0]
[0, 0, 0, 1, 0]]

+



One hot encoding

Score caluclated and confidence score check

- Deterministic Logic (Gate 1): Executes JSON Rule DSL locally on-device for offline functionality in zero-connectivity regions.
- Predictive ML Scorer (Gate 2): Ranks qualified schemes by approval probability using server-trained LightGBM/TFLite for on-device inference.
- Multivariate Inputs: Processes household demographics, district-level poverty priors, and historical worker success rates.
- Explainability: Surfaces SHAP-lite values on the CHW console to provide transparent reasoning for scheme rankings.
  - Input features: household profile fields (one-hot encoded) + district-level priors (local poverty rate, historical acceptance rate per scheme) + binary document availability flags + CHW local success rate
  - Output formula: Score = predicted_success × impact_weight × urgency_factor

PDF AUTO-FILL ENGINE

Details regarding commercial employment proposed to be taken up:- (a) (i) Name of organisation (firm or company or co-operative society, etc.). (ii) Brief nature of the organisation. (iii) Full address of the registered office of the organisation. (iv) Permanent Account Number or Tax Identification Number or Registration Number of the organisation. (b) Products being manufactured by the firm/type of business carried out by the firm etc. (c) Whether the officer had during the last three years of his official career, any dealings with the firm or company or cooperative society, etc. (d) Duration and nature of the Official dealing with the firm (e) Name of the job/post offered (f) Whether post was advertised, if not, how offer made (Attach Newspaper cutting of the advertisement and copy of the offer of appointment, if any) (g) Description of the duties of the Post/job, Remuneration offered for the post/job. (h) If proposing to set up practice, indicate: (i) Professional Qualification in the field of practice (ii) Nature of proposed practice	No. 142, 4th Cross, Kuvempuna 9845012378 876543219012 Widow 48000 KA/MYS/2019/0034521 OBC
Any other information which the Applicant desires to furnish in support of his request	36201234567890

01

\When a scheme is matched, the Form Rendering Service resolves this key to a MinIO object path and fetches the actual official government PDF.

02

Coordinate-Based Overlay Each template has a companion JSON field-map that specifies for every form field: which page it is on, its X/Y pixel coordinates, font size, and maximum character width.

03

Integrity Checks

- Template minimum file-size validation (catches stub/placeholder PDFs accidentally left in MinIO)
- Placeholder blocking (process errors and logs rather than producing a defective output)

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CHW VERIFICATION CONSOLE

- ZELIA treats all voice-extracted data as provisional. Only values confirmed from physical identity documents are treated as authoritative and used for the final eligibility decision.

extract textual data and save to PostgreSQL as metadata with key value pairs

Phase 1 Document Ingestion & OCR The CHW photographs or uploads the beneficiary's identity documents. The validation service then:

- Runs TFLite OCR to extract raw text from the document
- For Aadhaar specifically: decodes the embedded secure QR code and verifies its RSA digital signature against the UIDAI public key giving a cryptographic guarantee of authenticity

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CHW VERIFICATION CONSOLE

- **Phase 2 – Field-by-Field Comparison** Each extracted document field is compared to the corresponding value in the auto-filled form.
  - **VERIFIED –** document and form values match within tolerance (age is allowed ±1 year to account for birthday proximity during interview)
  - **MISMATCH** – values differ beyond tolerance; both values are displayed side-by-side on the CHW console so the worker can see exactly what conflicts
  - **MISSING** – no document of the required type was submitted, or the field could not be extracted reliably from the provided document

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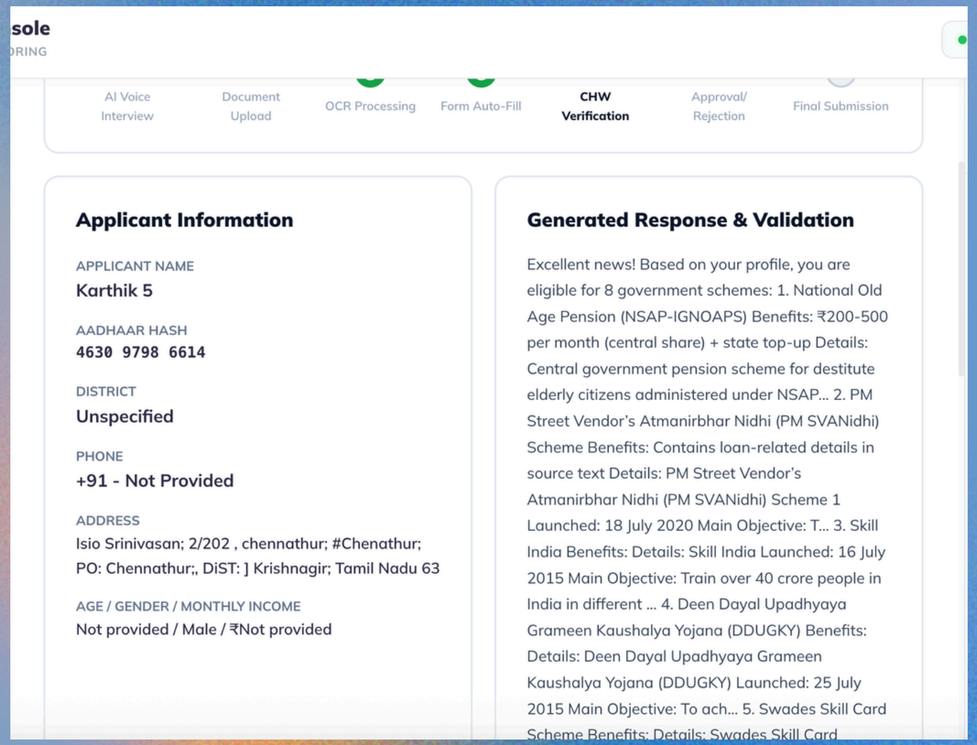
CHW VERIFICATION CONSOLE



Phase 3 – Eligibility Re-check with Authoritative Values

- After document fields are extracted, the system re-runs the eligibility rules engine using document values instead of voice values.

Phase 4 – Approval Lock Mechanism The Approve button on the CHW console remains disabled until all three conditions are simultaneously satisfied:

- All required documents for that scheme have been submitted
- Every form field that can be cross-checked with a document is marked VERIFIED
- The re-check with the authoritative document values returns eligible: true

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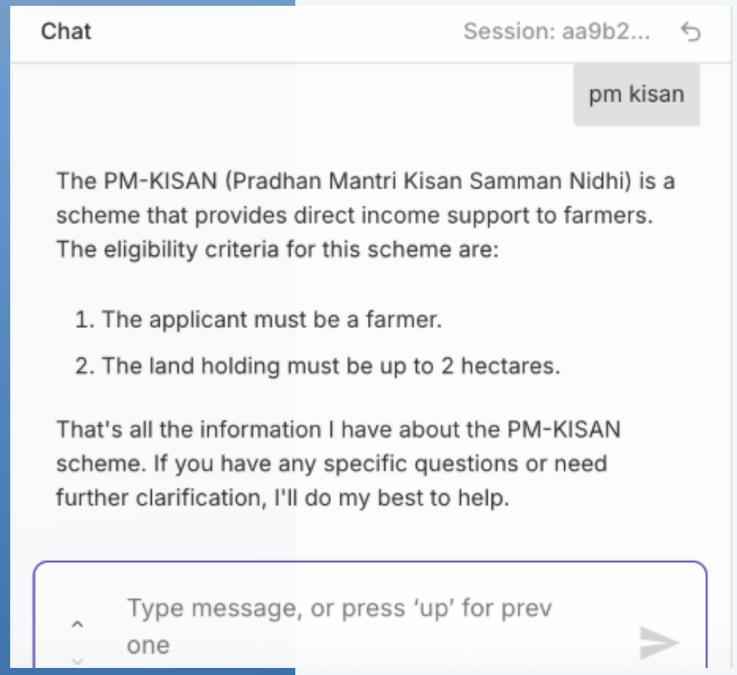


ZELIA COPILOT (RAG CHATBOT)

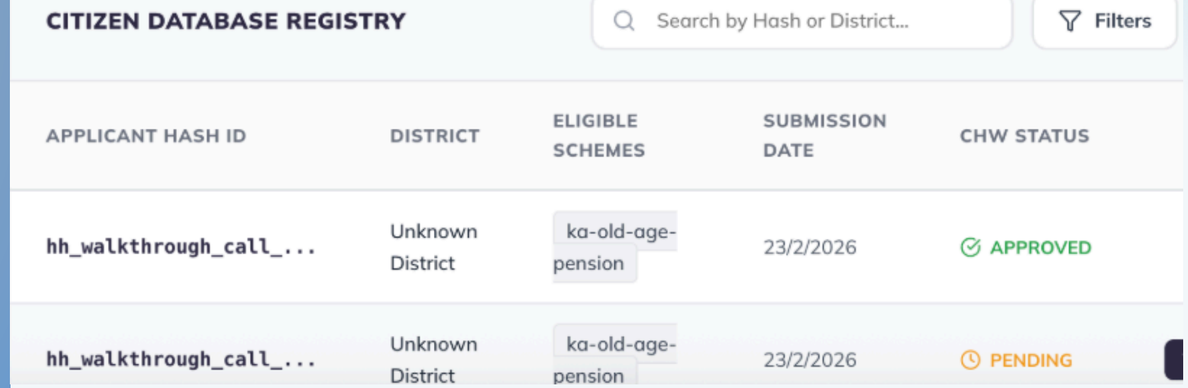
01 **Document Indexing (offline, one-time):** All government scheme documents, eligibility manuals, and FAQ pages are chunked into passages, embedded using Google Gemini Text Embeddings, and stored in Pinecone (title and description of each schemes are stored as vectors).

02 **Query Processing (live, per request):** The CHW types a natural-language question. The query is embedded and matched against the Pinecone index via maximum marginal relevance search. Top-k most relevant passages are retrieved.

03 **Answer Generation:** Retrieved passages are injected as grounded context into a prompt sent to Azure GPT-4o. The model generates a response strictly grounded in indexed scheme content — it cannot hallucinate policy details that aren't in the index.

04 **Response Delivery:** The answer is displayed in the persistent Copilot sidebar alongside the CHW's current workflow view.

SECURITY & PRIVACY



01

Full-disk AES-256 encryption for all application files and documents stored on the device

02

Consent is recorded as a signed audio clip at the start of every interview, with metadata hashed and timestamped in the local ledger creating a legally defensible consent record

03

Each household is identified by a salted SHA-256 hash. The salt is generated on-device and never uploaded, making it computationally infeasible to reverse the hash even with access to backend data

04

The HMAC key used to sign all ledger entries is derived from the device's hardware Trusted Platform Module (TPM) or Android Keystore, meaning it cannot be extracted by software.

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CONCLUSION

- A real-time pipeline that takes a raw voice transcript and produces a validated, filled government application form with zero human intervention in the data path
- A hybrid eligibility engine that combines zero-false-positive deterministic rules with ML ranking operating entirely offline on a low-end Android device
- A document-grounded verification model that ensures no application is approved on interview estimates alone — only on values confirmed from physical identity proofs
- A RAG chatbot that resolves colloquial, misspelled, multilingual queries to accurate scheme information grounded in indexed government documents
- A QA chain that produces reproducible, dated evidence artifacts: ASR WER gate results in three languages, a PDF audit with zero critical findings, and a production readiness report with 100% template coverage